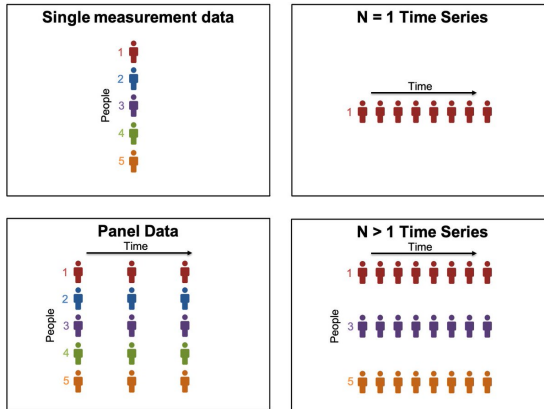


Advanced Econometrics

Introduction to Time Series

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What is Time Series Analysis?



A **time series** is data collected over time, in order.

Examples:

- Daily stock prices
- Monthly unemployment rate
- Weekly sales
- Hourly temperature

Main idea:

- The order matters
- Yesterday can affect today

What Do We Look For?

Important patterns in time series:

- **Trend:** long-term increase or decrease over time
 - Example: population growth, increasing sales
- **Seasonality:** regular repeating patterns at fixed intervals
 - Example: higher retail sales in December, more travel in summer
- **Noise:** random variation that cannot be explained
 - Example: unexpected shocks, measurement error

Why this matters:

- Helps us understand the structure of the data
- Improves forecasting accuracy
- Allows us to detect unusual events or changes

Example:

- Ice cream sales increase over years (**trend**)
- Sales spike every summer (**seasonality**)
- Daily fluctuations (**noise**)

What Can We Do With Time Series?

Main things we can do:

1 Describe patterns

- Identify trends, seasonality, and unusual behavior
- Understand how the data evolves over time

2 Forecast future values

- Use past data to predict future outcomes
- Important for planning and decision-making

3 Study effects of interventions

- Analyze what happens before and after an event
- Example: new policy, pricing change, or external shock

4 Compare relationships between variables

- Check whether one time series helps predict another
- Example: advertising and sales, weather and demand

Examples:

- Predict next month's sales (**forecasting**)
- Check whether a new policy changed crime (**interrupted time series**)
- Test whether advertising helps predict sales (**Granger causality**)

Prediction / Forecasting

Forecasting means predicting future values using past data.

Common methods:

- Moving average
- Exponential smoothing
- ARIMA
- Machine learning models

Example:

- Use past monthly sales to predict next month's sales

Very Simple Forecasting Example

Suppose weekly coffee sales are:

100, 105, 110, 108, 112

A simple forecast might use the recent pattern to guess the next value.

Maybe next week is around:

113 or 114

This is the basic forecasting idea:

Use the past to estimate the future

Interrupted Time Series (ITS)

Interrupted Time Series asks:

Did something change after an event or intervention?

Examples of interventions:

- New law
- New tax
- COVID lockdown
- New treatment in a hospital

Interrupted Time Series: What Changes?

In ITS, we often look for:

- **Level change:** sudden jump or drop
- **Slope change:** trend changes after intervention

Example:

- Before a sugar tax: soda sales stable
- After tax: soda sales drop

Then we ask:

Was the intervention associated with that change?

Interrupted Time Series Example

Example: monthly hospital infections

- Jan–Jun: infection rate around the same
- July: new hygiene policy introduced
- Aug–Dec: infection rate falls

Interpretation:

- The policy may have reduced infections
- ITS helps measure that before/after effect

Autocorrelation

Autocorrelation means a series is related to its own past values.

Question:

Is today's value similar to yesterday's value?

Examples:

- Temperature: usually yes
- Sales: often yes
- Lottery numbers: no

If autocorrelation is strong, the past helps prediction.

Stationarity

A time series is **stationary** if its statistical properties stay stable over time.

Usually this means:

- **Constant mean:** no long-term upward or downward trend
- **Constant variance:** fluctuations are roughly the same size
- **Stable structure:** relationships over time do not change

Intuition:

- The series behaves the same today as it did in the past
- No systematic growth, decline, or changing volatility

Why do we care?

- Many models (e.g. ARIMA) assume stationarity
- Easier to model and predict stable data
- Non-stationary data can give misleading results

What if data is not stationary?

- Remove trend (detrending)
- Difference the data (use changes instead of levels)
- Transform (e.g. log transformation)

Differencing (Making Data Stationary)

Differencing removes trends by looking at changes between consecutive values.

First difference:

$$y'_t = y_t - y_{t-1}$$

Intuition:

- Instead of modeling the level, we model the **change**
- Removes upward or downward trends

Example:

- Original data: 100, 105, 110, 115
- First difference: 5, 5, 5

Types of differencing:

- **First difference:** removes linear trend
- **Second difference:** removes more complex trends

$$y''_t = y'_t - y'_{t-1}$$

Why it helps:

- Makes the series more stable (stationary)
- Allows models like ARIMA to work properly

Levels vs Differences

We can analyze a time series in:

- **Levels:** the original values
- **Differences:** changes from one period to the next

Why does this matter?

- Levels may contain trends
- Differences are often more stationary

Example:

- Level: monthly price
- Difference: monthly price change

What Is a Lag?

A **lag** is a past value of a time series.

Examples:

- y_{t-1} = value one period ago
- y_{t-2} = value two periods ago

Why lags matter:

- The past often helps explain the present
- Many time series models use lagged values

Example:

$$\text{Sales}_t = \alpha + \beta \text{Sales}_{t-1} + u_t$$

Decomposition

We can split a time series into parts:

$$\text{Data} = \text{Trend} + \text{Seasonality} + \text{Noise}$$

Why useful?

- Helps us understand the data
- Makes forecasting easier

Example:

- Airline passengers increase over years
- Passenger counts peak every summer

Granger Causality

Granger causality does **not** prove true causation.

It asks:

Do past values of X improve prediction of Y?

If yes, then:

X Granger-causes Y

Granger Causality Example

Example:

- X = advertising spending
- Y = sales

Question:

Do past advertising values help predict future sales?

If yes:

- Advertising Granger-causes sales

But careful:

- This is about **prediction**
- It is not proof of real-world causation

Why Granger Causality Can Be Tricky

Even if X Granger-causes Y:

- There may be hidden variables
- Both X and Y may be driven by something else

Example:

- Ice cream sales predict drowning accidents
- But summer weather affects both

So:

Granger causality = useful predictive relationship, not final proof

Spurious Regression: What Is It?

Definition:

- A regression that shows a statistically significant relationship
- Even though the variables are not meaningfully related

Why it happens:

- Non-stationary (trending) time series
- Common underlying trends
- Pure coincidence

Looks real in data, but is actually misleading

Spurious Regression: Non-Stationary Data

Problem:

- Many time series have **unit roots** (they are non-stationary)
- Their mean and variance change over time

What happens:

- Regressing one non-stationary series on another
- Often produces:
 - High R^2
 - Significant coefficients

Result: misleading relationship driven by trends

Spurious Regression: Common Trends

Problem:

- Two variables follow a similar upward or downward trend over time

Example:

- GDP $\uparrow$ over time
- CO₂ emissions $\uparrow$ over time

What happens:

- Regression detects a strong relationship
- But both are driven by a **shared time trend**

Trend $\neq$ true economic relationship

Spurious Regression: Omitted Variables

True model:

$$Y = \beta_0 + \beta_1 X + \beta_2 Z + u$$

But we estimate (omitting Z):

$$Y = \alpha_0 + \alpha_1 X + \varepsilon$$

Problem:

- If Z affects both X and Y
- Then α_1 is biased:

$$\alpha_1 = \beta_1 + \beta_2 \cdot \frac{\text{Cov}(X, Z)}{\text{Var}(X)}$$

Estimated effect mixes true effect + omitted variable bias

Famous Nonsense Correlations

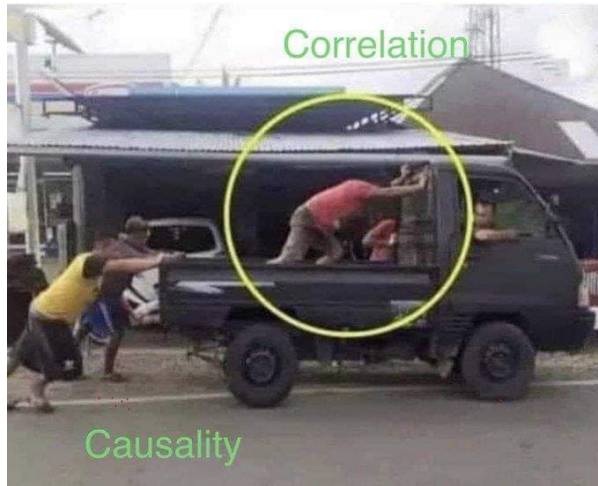
- Number of pirates vs global temperature
- Cheese consumption vs deaths by bedsheet entanglement
- Nicolas Cage movies vs swimming pool drownings

Key insight:

- With enough data, random variables can appear correlated

Correlation $\neq$ Causation





Cointegration: When Non-Stationary Series Move Together

Important exception:

- Two variables can each be non-stationary
- But still have a meaningful long-run relationship

This happens when a linear combination is stationary:

$$Y_t - \beta X_t \sim I(0)$$

Interpretation:

- The variables may drift over time
- But they do not drift too far apart
- They are linked by a long-run equilibrium

Non-stationary + cointegrated $\neq$ spurious regression

Thank you